Modulation Discovery with Differentiable Digital Signal Processing

centre for digital music

Christopher Mitcheltree [♭] Hao Hao Tan [♯] Joshua D. Reiss [♭]

[♭] Centre for Digital Music, Queen Mary, University of London
[♯] Independent Researcher, Singapore

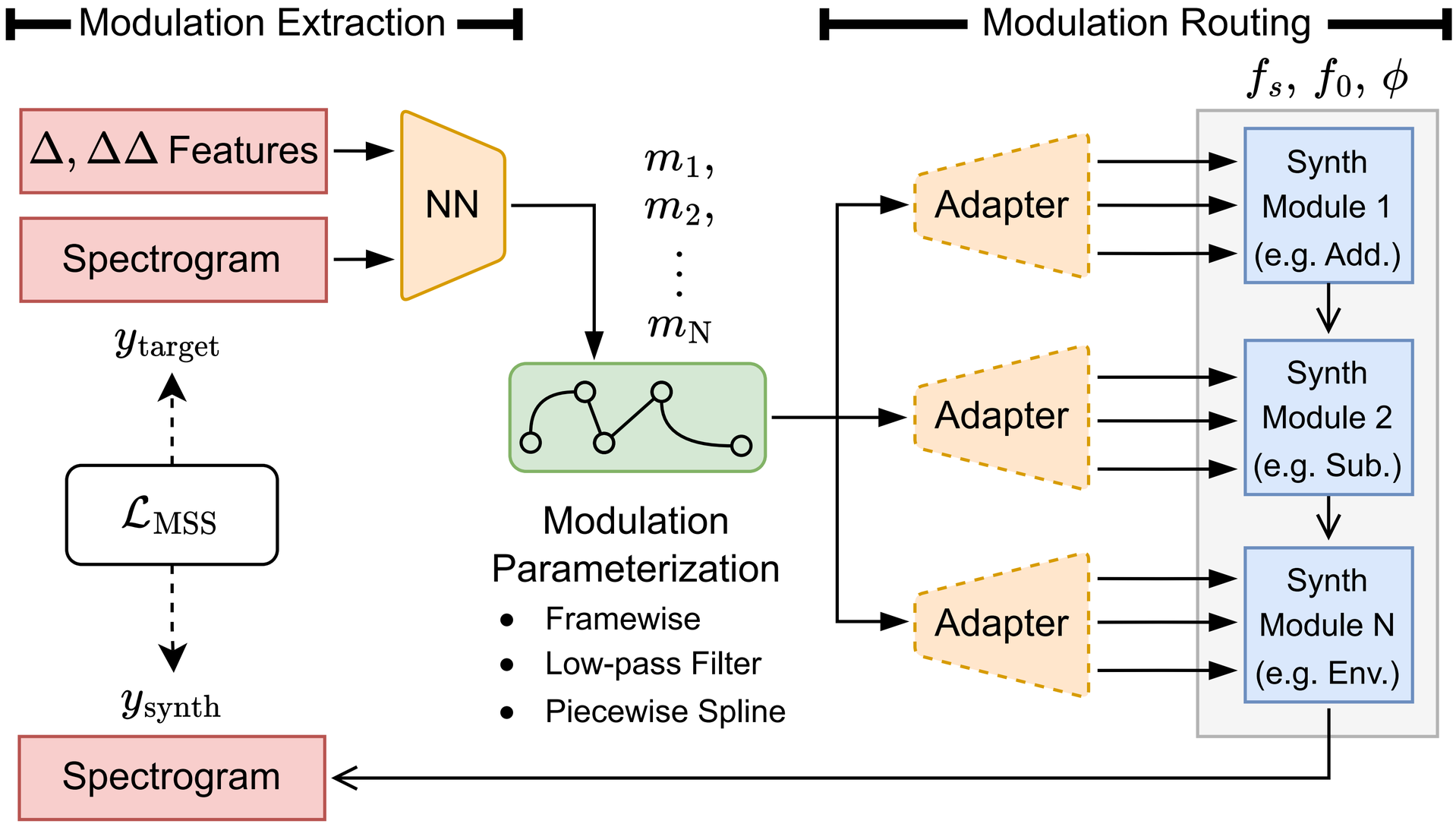

Figure 1. Overview of the modulation discovery process through modulation extraction, parameterization, and routing using a DDSP synth. Orange blocks are neural networks, dashed blocks are optional, and blue blocks are differentiable and may contain learnable weights for sound matching.

Overview

- **Motivation:** Modulation is central to sound design and music production — virtually all modern synthesizer presets use it to create dynamic, evolving audio.
- **Problem:** Existing sound matching systems act as black boxes or predict uninterpretable, high-dimensional framewise modulation signals without considering modulation routing and the human-centric sound design process.
- **Approach:** We introduce a modulation discovery framework combining modulation extraction, constrained curve parameterizations, and modulation routing through differentiable digital signal processing (DDSP) and self-supervised sound matching.
- **Experiments:** We demonstrate its effectiveness on synthetic and real-world audio, recovering interpretable mod. signals across multiple DDSP synth architectures.
- **Findings:** Low-pass filtering provides the best balance between sound matching accuracy and interpretability, while piecewise splines maximize interpretability.
- **Contributions:** We make our code, audio listening samples, and the trained DDSP synths available via the accompanying website and the Neutone FX plugin.

Figure 2. A 2D drawable modulation grid in the `Vital` soft synth.

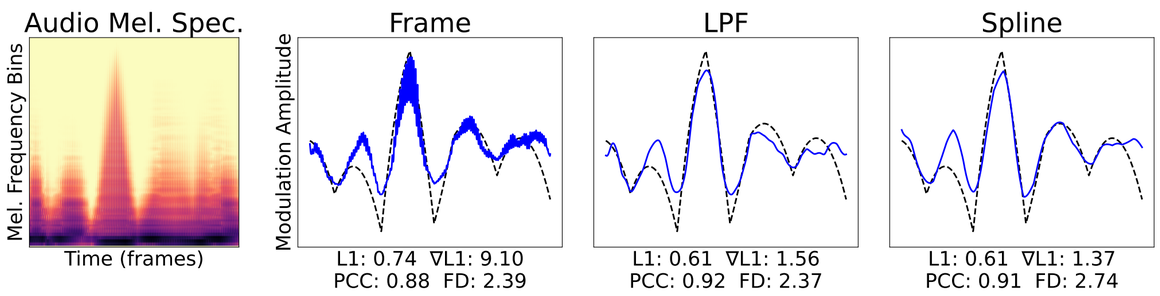

Figure 3. Modulation extraction with the framewise (Frame), low-pass filtered (LPF), and piecewise Bézier curve (Spline) parameterizations. The ground truth signal is dashed and four modulation signal distance measures are computed (see Section 3.1 in the paper).

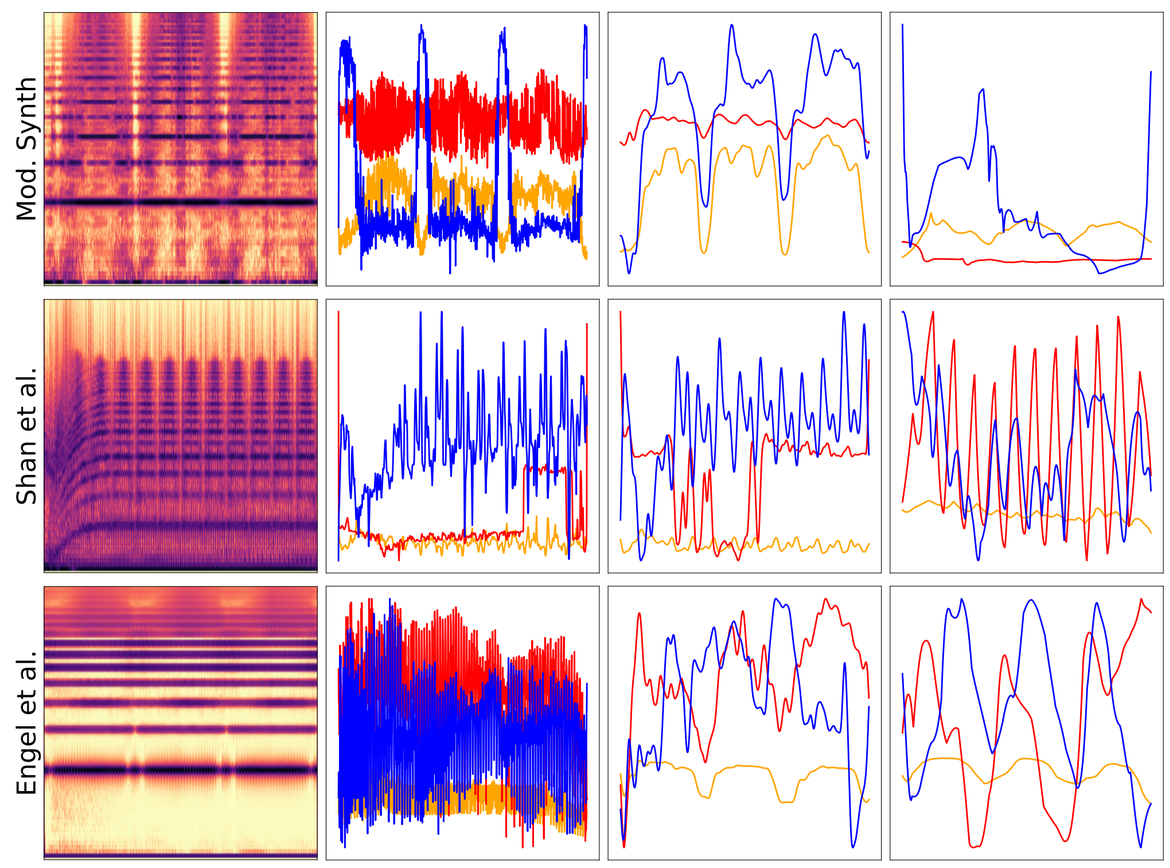

Figure 4. Discovered additive (red), subtractive (blue), and envelope (orange) modulations in `Serum` test dataset audio samples using different DDSP synths and Frame (left), LPF (center), and Spline (right) modulation signal parameterizations.

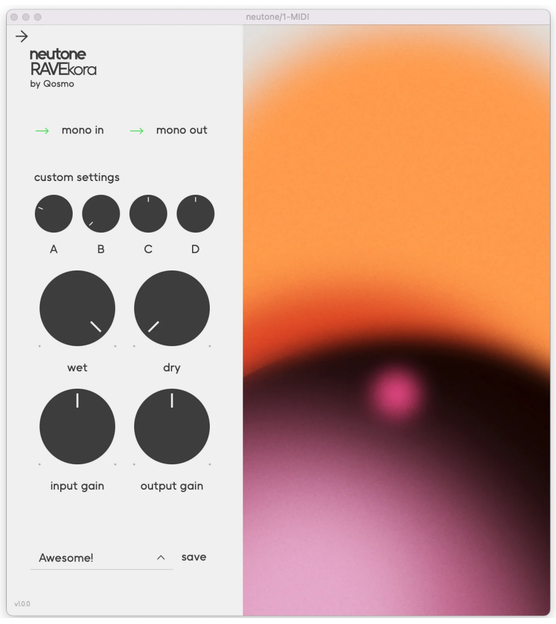

Figure 5. Learned DDSP synths are provided as plugins through the Neutone SDK.

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